Connecting neuroimaging knowledge graphs across data, publications and labs

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Slides, code, graphs and provenance
github.com/ReproNim/llm-workshop-kg

How each slide was made

Production

- **H** written by the speaker
- **L** drafted by an LLM, edited by the speaker
- **S** an agent followed a named skill (named on slide)
- **A** general agent work: own code, no skill
- **R** retrieved by an agent from cited sources
- **M** answered by an external agent via MCP
- **P** produced earlier by a different agentic system

Checking

re-run
recomputed independently

validated
passed automated checks

reviewed
read and corrected by the speaker

unverified
not independently checked

Wikipedia and Wikidata: prose and a structured graph

In the graph: Not our graph: Wikidata's public graph (query.wikidata.org SPARQL); same pattern as the graphs in this talk

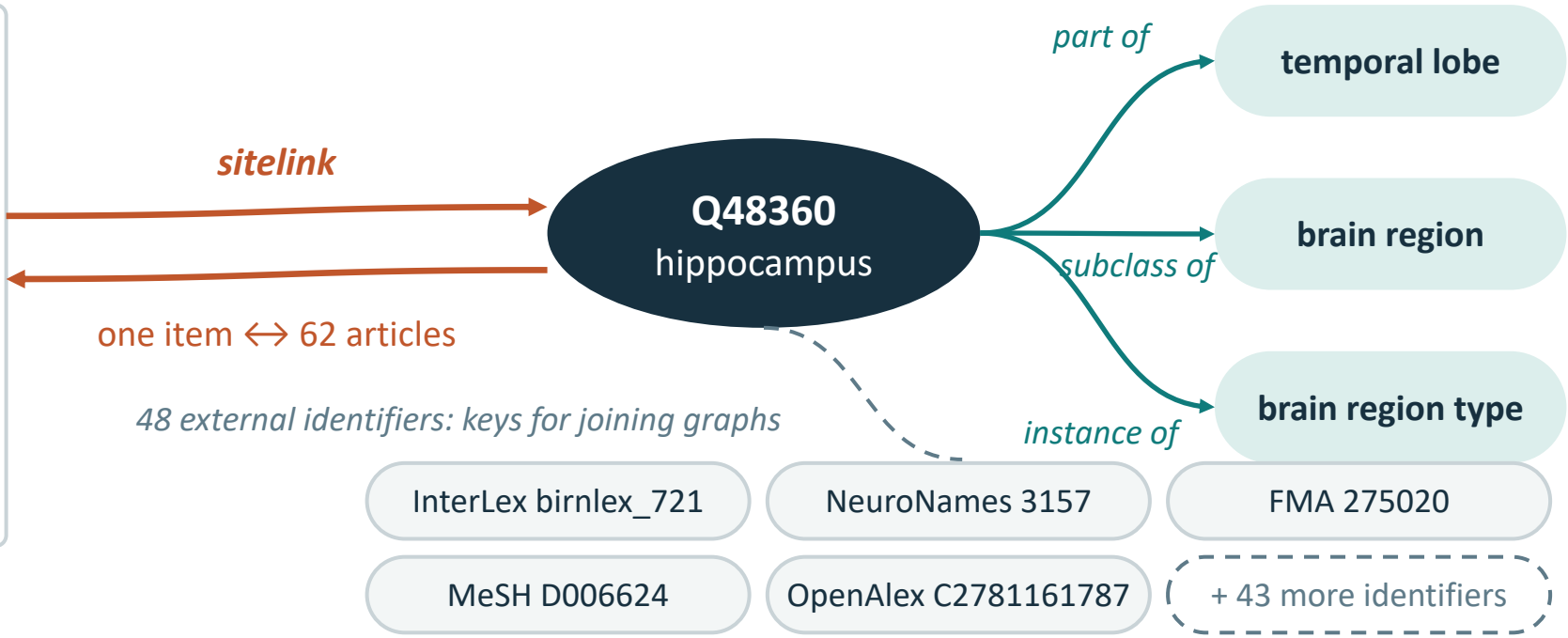
Wikipedia: written for people

Hippocampus

"... a major component of the brain of humans and many other vertebrates ..."

62 language versions

Wikidata: statements a program can use



Query: `?region –part of→ temporal lobe`

→ 5 recorded: archicortex, auditory cortex, entorhinal cortex, hippocampus, and one unlabelled item

returns only what someone recorded · Wikidata has no UBERON ID for the hippocampus, so a UBERON join would miss it

R A *validated*

Data: Wikidata Q48360 (rev. 2516828778) and English Wikipedia 'Hippocampus' (rev. 1374836959), retrieved 2026-10-16

Learning objectives

By the end of this talk, you should be able to:



Represent a neuroimaging result as a knowledge graph

linking the research question, the claim and its scope, the participants and derived measures it relies on, and the source data



Connect information across knowledge graphs

joining your data with public datasets, publications, tools and collaborators' work through shared identifiers and vocabularies, and deciding what can leave your lab

The two example datasets

In the graph: Counts read by SPARQL: ds000030 graph (combined_graph.ttl, 1,113,959 triples); ABIDE graph/core (queries/Q14.rq)

ds000030

UCLA Consortium for Neuropsychiatric Phenomics · OpenNeuro, commit 4070b6e, CC0

- 272 adults: 130 controls, 50 schizophrenia, 49 bipolar disorder, 43 ADHD
- structural MRI, resting and task fMRI (e.g. stop-signal)
- 52 questionnaire and phenotype tables

Papers used here

Poldrack et al. 2016: data descriptor
Gorgolewski et al. 2017: preprocessed derivatives
Kim et al. 2023: prediction across UCLA and COBRE

Used in Part 1: representing a result

ABIDE I

Autism Brain Imaging Data Exchange · ABIDE Preprocessed (C-PAC), CC BY-NC-SA

- 1,112 participants (539 autistic, 573 typically developing) from 17 sites (20 site codes in the data)
- resting-state fMRI; phenotypes incl. IQ, ADOS, SRS
- 200-region time series for 1,035 participants

Papers used here

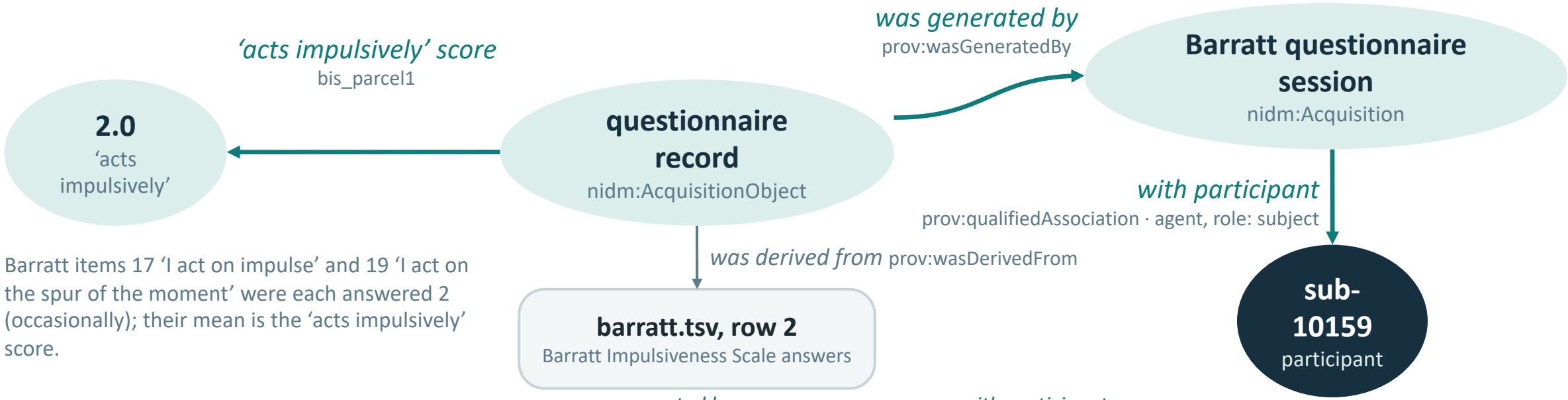
Di Martino et al. 2014: group differences
Abraham et al. 2017: site-held-out prediction
a 74-study literature review

Used in Part 2 and the Claude Code examples

Data: ds000030 (OpenNeuro, commit 4070b6e); ABIDE I (ABIDE Preprocessed)

A knowledge graph in four statements

In the graph: Path query over ds000030 nidm_phenotypes.ttl: record → bis_parcel1; wasGeneratedBy → session → qualifiedAssociation/agent → participant: 272 rows



A query



Triple
one statement: subject, relation, object

Identifier
a name that means one thing everywhere (sub-10159, a DOI)

Vocabulary
agreed names for relations and types (NIDM, PROV)

Query
a statement pattern with blanks (SPARQL)

Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

Components of a dataset and a publication

In the graph: Dataset side: ds000030 graph and ABIDE graph/core · publication side: claims_and_questions.ttl, literature_kg.ttl (17 / 74 reconstructible)

Dataset

Participants and phenotypes
272 participants; 52 tables, 1,983 columns

Data dictionary
'acts impulsively' = mean of Barratt items 17 and 19

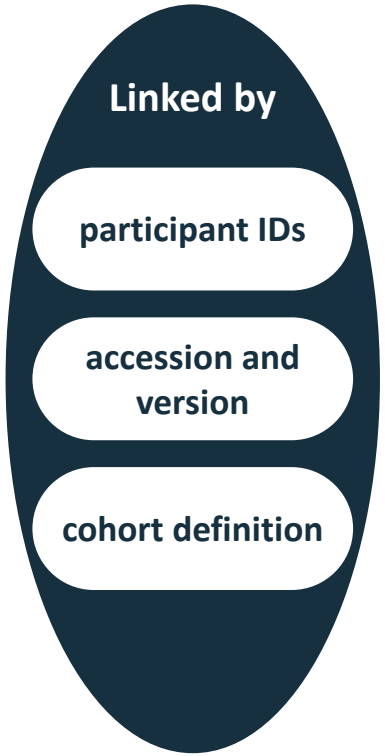
Scan sidecars
sub-10159 BOLD: TR 2 s, TE 0.03 s, 3 T, scanner 35343

Task events
stop-signal task: 96 go, 32 stop trials, 128 rest intervals

File inventory
2,531 file pointers

Derivatives
questionnaire scores; 200-region time series (C-PAC)

Release and version
ds000030 commit 4070b6e; ABIDE Preprocessed



Publication

Research question
Kim 2023: which direction of connectivity change is associated with schizophrenia?

Cohort and exclusions
Kim 2023: 47 patients + 47 controls; Abraham 2017: 871 of 1,112

Preprocessing and versions
fMRIPrep: fMRI to anatomy (FreeSurfer 6.0.0), anatomy to MNI template (ANTs 2.1.0)

Analysis and evaluation
linear classifier; leave-one-participant-out, leave-site-out

Results and claims, with scope
both directions altered; evidence 70.2% accuracy, UCLA, leave-one-out

Data and code availability
cohorts reconstructible for 17 of 74 ABIDE studies (our review)

P R A *re-run · validated*

Data: ds000030 (UCLA CNP, commit 4070b6e); ABIDE I (ABIDE Preprocessed, C-PAC); 74-study review (2026-10-15)

Building knowledge graphs from datasets and papers

In the graph: Sandbox page names read from the brainkb-ui source (commit a4815a3), not exercised in this session; skills in sensein/agent_skills

A dataset
BIDS + phenotype tables

→ **neuroimaging-kg skill (agent_skills PR #49)**
PyNIDM bidsmri2nidm / csv2nidm; ReproNim NIDM BIDS apps (FreeSurfer, ANTs, MRIQC → NIDM)

→ **NIDM / PROV dataset graph**

Used here: *ABIDE graph by general sub-agents, no skill; this skill was written from that work afterwards → Claude Code examples*

One publication
PDF or DOI

→ **StructSense skill**
sandbox: Neuroscientific Named Entity Extraction; Structured Resource Extraction (BrainKB Resource Ontology)

→ **claims, entities, datasets and tools, with evidence spans**

Used here: *Kim 2023 claims from a StructSense skill pilot, run by Codex*

A body of publications
a topic or question

→ **SynthScholar skill**
sandbox: SynthScholar — protocol → search strategy → screening and charting

→ **review graph (SLR ontology)**

Used here: *74 ABIDE studies: SynthScholar skill for protocol, retrieval, export; screening and charting by sub-agents*

Ingest into BrainKB
sandbox: Ingest Knowledge Graph, then job status

Query it
brainkb skill · BrainKB MCP

Questionnaires: sandbox ReproSchema Extraction (PDF → ReproSchema assessments)

sandbox.brainkb.org

Data: ABIDE I; Kim et al. 2023; 74 ABIDE studies

Entities in a neuroimaging knowledge graph

In the graph: Node and edge types; teal = types used in our ds000030 and ABIDE graphs (assigned in the plan, not computed by a query)

What was studied

- Datasets, versions
- Participants
- Acquisitions
- Tasks, constructs

How it was done

- Methods, software
- How to apply
- Templates, spaces
- Run provenance
- Compute envs



Who and in what context

- Publications
- People, roles
- Projects, funding
- Reviews, disputes
- Resource activity
- Ontologies
- Consent, licences

What was found

- Derived measures
- Statistical maps
- Questions
- Scoped claims

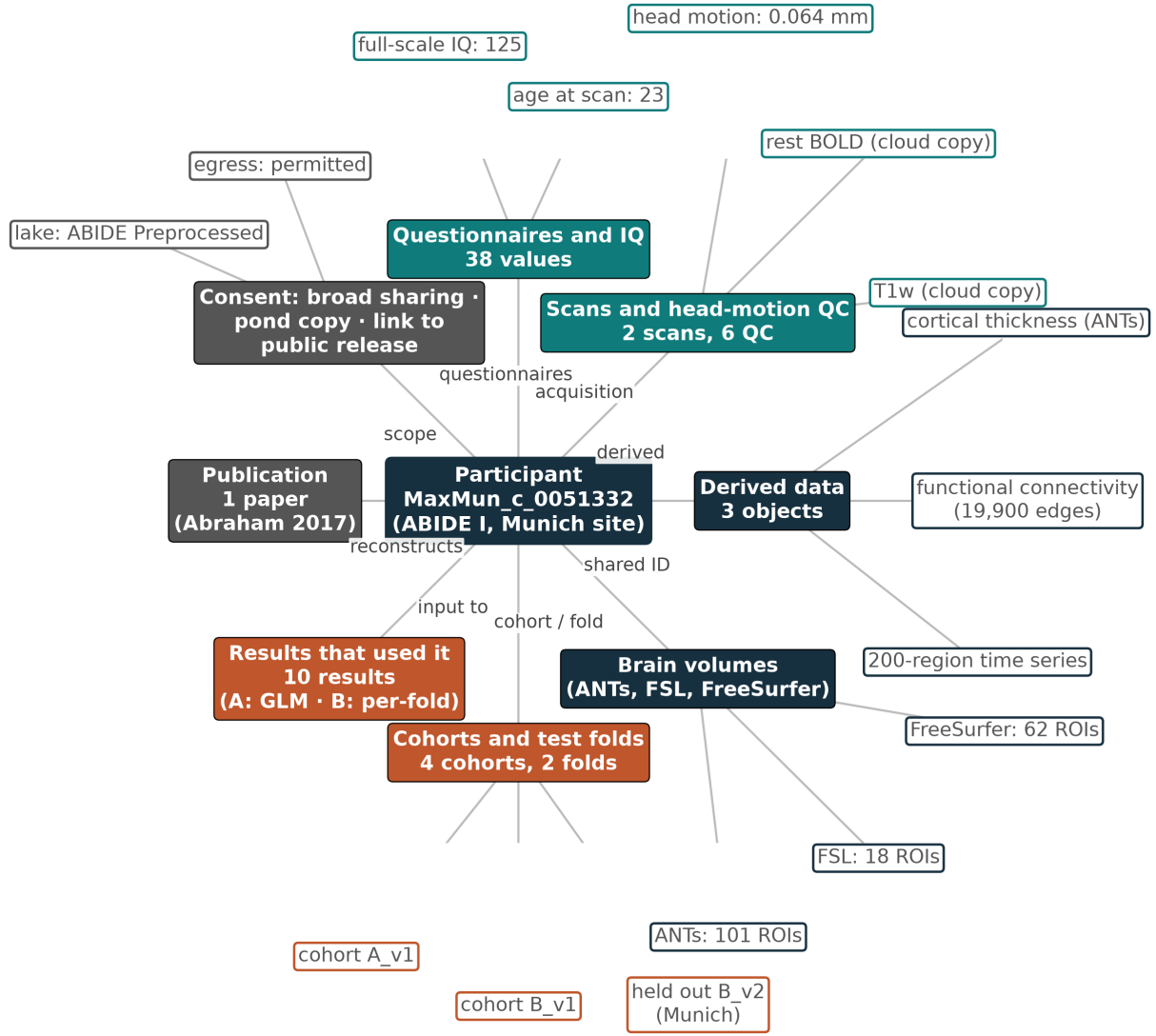
teal = in the worked examples

Which of these does your lab already write down?

H L R *reviewed*

Knowledge graph around a single participant

In the graph: 13 SPARQL queries (queries/Q13_*.rq) over graph/core, scope, view/volumes → 239 triples → figure: code/participant_ego.py



One ABIDE participant, MaxMun_c_0051332
everything within two steps: 239 statements, one query (0.03 s)

- **Phenotype** 38 values
- **Scans and QC** 2 scans; head motion 0.064 mm
- **Derived data** regional time series, connectivity
- **Brain volumes by tool** ANTs 101 · FSL 18 · FreeSurfer 62 values
- **Cohorts and folds** in 4 cohorts; out of A's stricter one (site too small)
- **Results** input to 10 results
- **Publication** Abraham 2017 cohort
- **Consent scope** broad sharing permitted

A *re-run · validated*

Data: ABIDE I (ABIDE Preprocessed, C-PAC); ReproNim simple2 NIDM (commit e0dff17)

Linking a claim to its evaluation and cohort

In the graph: claims_and_questions.ttl (StructSense skill pilot, run by Codex): question → claim → evaluation → cohort; evidence re-run by kim2023_rerun/rerun_ucla_cpm.py

Question · Kim 2023
Does increased connectivity add information about schizophrenia?

resting-state fMRI · 268-region atlas · patients vs controls

Claim
Yes: increased + decreased together classify best

Evidence
66 of 94 correct (70.2%); decreased alone 52%

Cohort
47 + 47, matched for age, sex, scanner

Leave-one-out
train on 93, predict the 94th, repeat for all; chance = 50%

Re-run here
authors' OSF data and code give 70.2% again; also 52.1% decreased alone, 61.7% increased alone (not in the paper)

Data
ds000030 (UCLA CNP), versioned

?

Partial link: authors' OSF data + code reproduce 66/94; not given: which ds000030 participant each matrix is

P S R A *re-run · validated*

Data: ds000030 (UCLA CNP, commit 4070b6e); Kim et al. 2023 (UCLA cohort from ds000030)

Exercise: evaluating claims against a source

In the graph: Statement A is a claim record with its evidence span (claims_and_questions.ttl, StructSense skill pilot); B and C were written for this exercise, not in the graph

A

"Each UCLA participant left out once: 66 of 94 (70.2%) classified correctly."

B

"Connectivity predicts schizophrenia with 70.2% accuracy in any cohort."

C

"70.2% proves a causal disease mechanism."

All three cite the same quote from Kim 2023. Which would you accept?

Data: Kim et al. 2023 (UCLA cohort from ds000030)

Quotation checks and scientific entailment

In the graph: StructSense skill's span and schema checks on the extracted records (results/skill/pilot_report.json); entailment is a human judgement, not in the graph

A

"Each UCLA participant left out once: 66 of 94 (70.2%) classified correctly."

Supported, in scope

B

"Connectivity predicts schizophrenia with 70.2% accuracy in any cohort."

Scope dropped

C

"70.2% proves a causal disease mechanism."

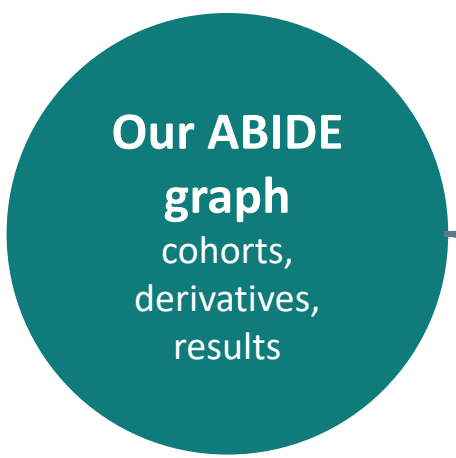
Not in the source

All three quote the same span, and an exact-quote check passes all three: it confirms the words are in the source, not that the statement follows from them. C passed the StructSense skill's span validator; a human reading rejected it.

Data: Kim et al. 2023 (UCLA cohort from ds000030)

Joining graphs through shared identifiers

In the graph: queries/simple2_identity.rq over graph/link + 27 simple2 site graphs; graph/publications.ttl (Q15.rq); literature_kg.ttl; Brain Researcher via MCP, not in our store



ReproNim simple2 NIDM ReproNim's ABIDE records, 27 sites *key: participant ID*
1,112 / 1,112 matched, none left over · 1,035 with time series · 767 in B's cohort (reviewed link, not declared identical)

AbstractAtlas OHBM 2026 abstracts *key: dataset name in abstract text*
35 abstracts · 17 state data use · 15 / 17 correct on hand check

OpenAlex open catalogue of papers *key: DOI and citation*
598 works cite Di Martino 2014 and name ABIDE

Brain Researcher KG an external AI assistant's graph *key: dataset node*
one aggregate ABIDE node (2,226 subjects); neither paper present

Literature review systematic review (SynthScholar skill + sub-agents) *key: DOI and cohort description*
9 of 74 reviewed studies reuse the 871-participant ABIDE I subset, incl. Abraham 2017, Parisot 2018, Ktena 2017

Same term, different measure

Cortical grey matter (UBERON:0000956; UBERON = anatomy vocabulary)

ANTs vs FreeSurfer: r 0.85, ratio 1.01

ANTs vs FSL: r 0.72, ratio 0.84

CSF (UBERON:0001359)

FSL total CSF vs ANTs ventricles: ~200× apart, r 0.23

A R S *re-run · validated*

Data: ABIDE I; simple2 NIDM (e0dff17); AbstractAtlas (OHBM 2026); OpenAlex; Brain Researcher KG; 74-study review (2026-10-15)

Harmonising graphs built by different people

In the graph: One named graph per source + PROV (code/store.py); SHACL code/shapes.ttl; LinkML check · Caltech numbers: neuroimaging-kg skill, compare_graphs.py

Our ABIDE graph
contributor A

1 Shared identifiers
dataset + version, participant ID, DOI, ORCID, RRID · Caltech, ours vs ReproNim simple2: 38 / 38 participants matched

2 Shared vocabularies
concept tags (InterLex, nidm-terms), UBERON regions, Cognitive Atlas tasks · Caltech: 12 concepts shared

3 Shared schema and validation
PyNIDM LinkML schema → SHACL rules; the same checks for every contributor

4 Reviewed mapping links
Caltech: 28 variables (over half ADOS, ADI-R or SRS subscores) share a source variable but not a concept → listed for review as skos:exactMatch candidates, never auto-asserted

5 One named graph per contributor, with PROV
Caltech: 0 of 282 participant × concept values conflict (both from the same ABIDE table) · across tools, 'CSF' is total CSF in FSL, ventricles in ANTs/FreeSurfer (~200×)

6 Federation instead of copying
Neurobagel sends the query to each pond; only permitted results return

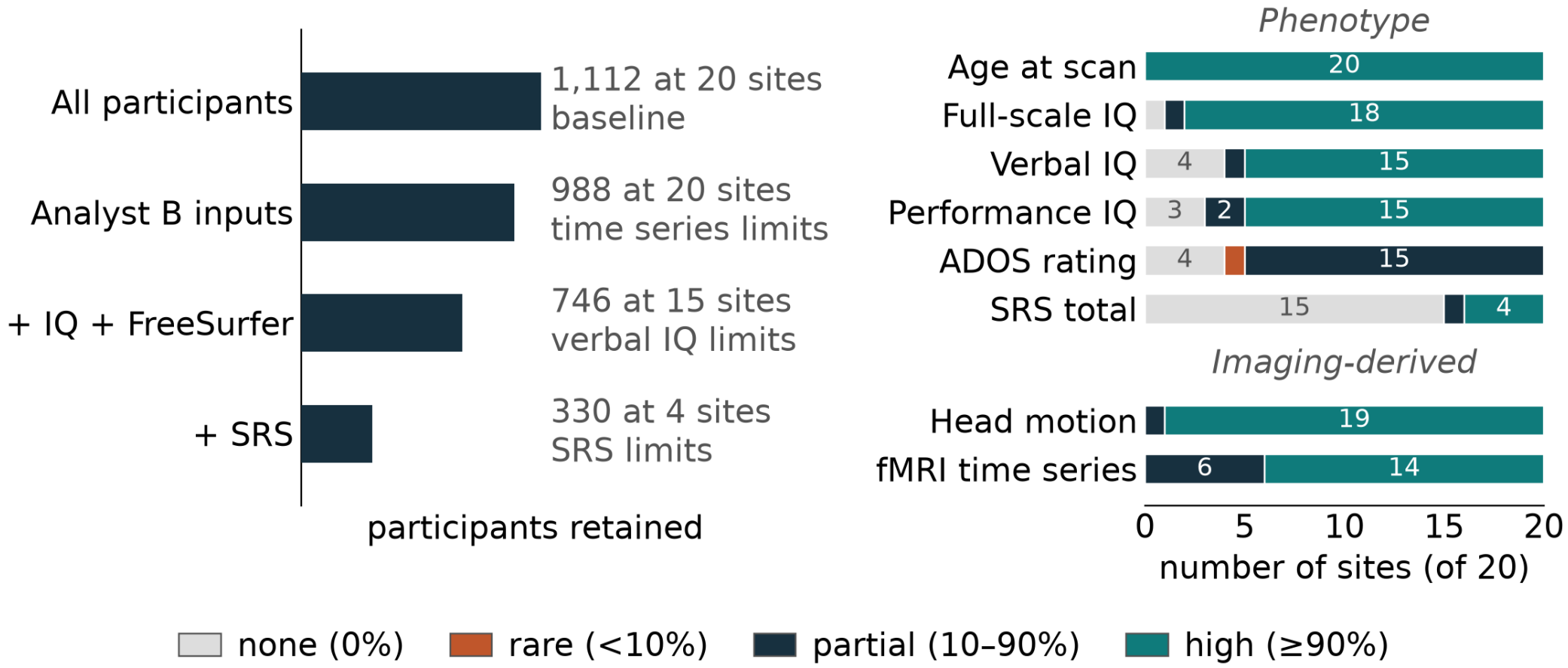
ReproNim simple2
contributor B

Data: ABIDE I (ABIDE Preprocessed, C-PAC); ReproNim simple2 NIDM (commit e0dff17)

Analyses supported by the joined data

In the graph: SPARQL (queries/Q14.rq) over graph/core + view/volumes + ReproNim simple2 graphs → counts by site and diagnosis (code/element_coverage.py → results/element_coverage.json) → figure: figs_slide/make_slide_figs.py

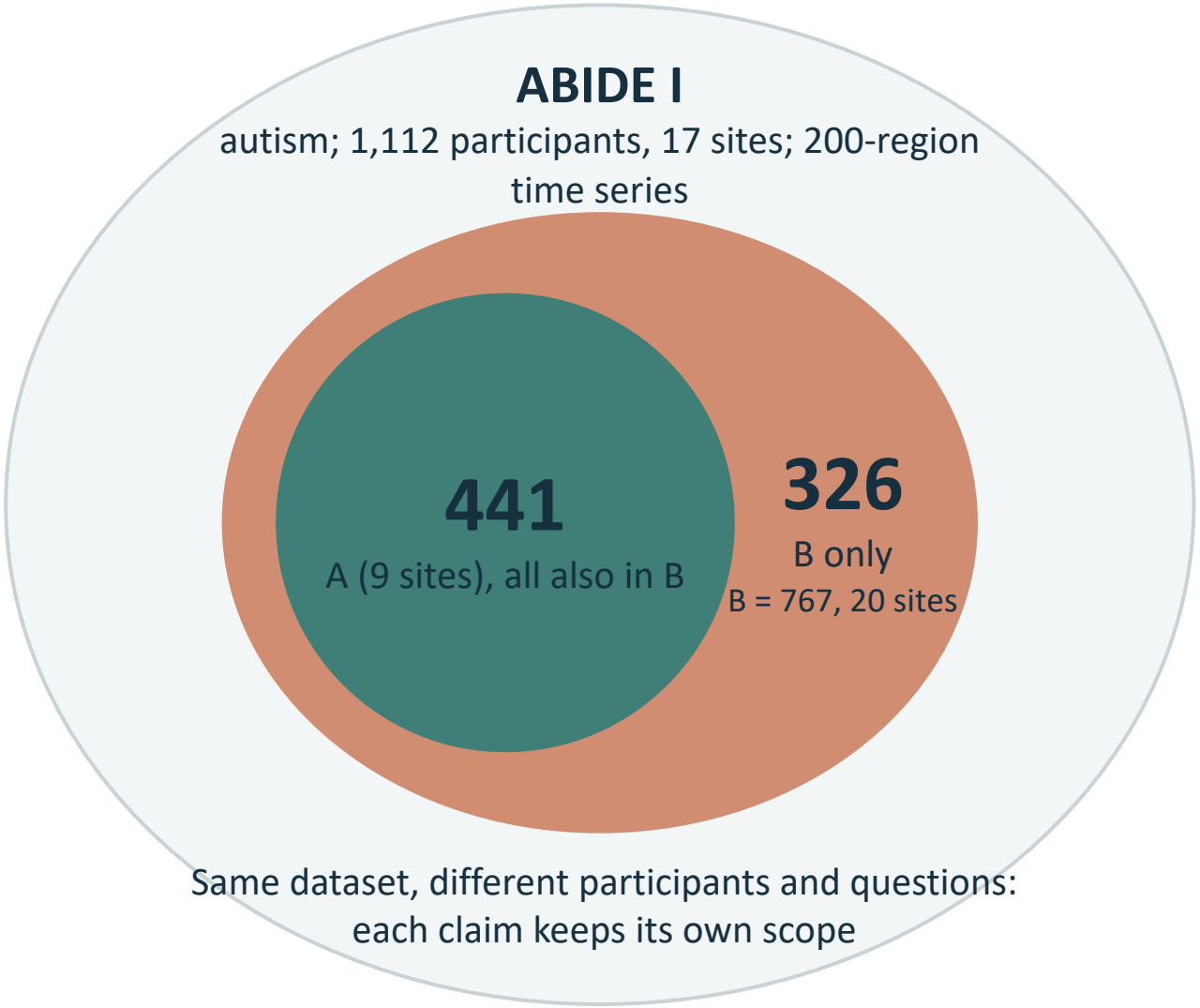
Left: participants left as each analysis adds a required element, and what limits it. Right: sites holding each element.



Data: ABIDE I (ABIDE Preprocessed, C-PAC); ReproNim simple2 NIDM (commit e0dff17)

ABIDE I: two analyses, two research questions

In the graph: Cohorts are prov:hadMember sets in graph/core (built by code/cohorts.py); overlap by queries/Q2.rq



Analyst A (simulated agent)

Do autistic and non-autistic groups differ in connectivity?

group difference · association

after Di Martino et al. 2014

Analyst B (simulated agent)

Does connectivity predict diagnosis at unseen sites?

prediction · leave-site-out

after Abraham et al. 2017

A R *re-run · validated*

Data: ABIDE I (ABIDE Preprocessed, C-PAC)

ABIDE I: group differences and site-held-out prediction

In the graph: Result, claim and review nodes in graph/core (code/analyst_a.py, analyst_b.py, reviews.py); reviews read by queries/Q6.rq, Q7.rq

Analyst A

5,428 connections differ

FDR-corrected (false discovery rate); 99.6% weaker in ASD (95% CI 90.2–99.8%)
52% of the weaker connections link the two hemispheres
median effect size d = 0.30 (difference in SD units) · n = 441, 9 sites
Caveat: the pattern tracks whole-brain mean connectivity (r = 0.98), so it may be a global difference
Reviewed by B: not established in held-out participants (pooled d −0.08, −0.31 to 0.15)

Association in this cohort

Analyst B

AUC 0.74 (0.70–0.77)

AUC: 0.5 = chance, 1 = perfect · tangent-space connectivity · tested on held-out sites, n = 767, 20 sites
correlation 0.72 · demographic + head motion 0.59 · demographic only 0.48
accuracy 66.9%; Abraham 2017 reported 66.8% on its own 871-participant cohort
Reviewed by A: top edges overlap A's 90 vs 54 expected (p ≈ 4e−8)

Prediction at held-out sites

A R *re-run · validated*

Data: ABIDE I (ABIDE Preprocessed, C-PAC)

Exercise: consequences of a stricter motion threshold

In the graph: The motion rule is stored on each cohort definition in graph/core (ex:threshold 0.2 for v1, 0.15 for v2)

Analyst A tightens the head-motion rule: mean FD (average movement) < 0.2 mm → < 0.15 mm

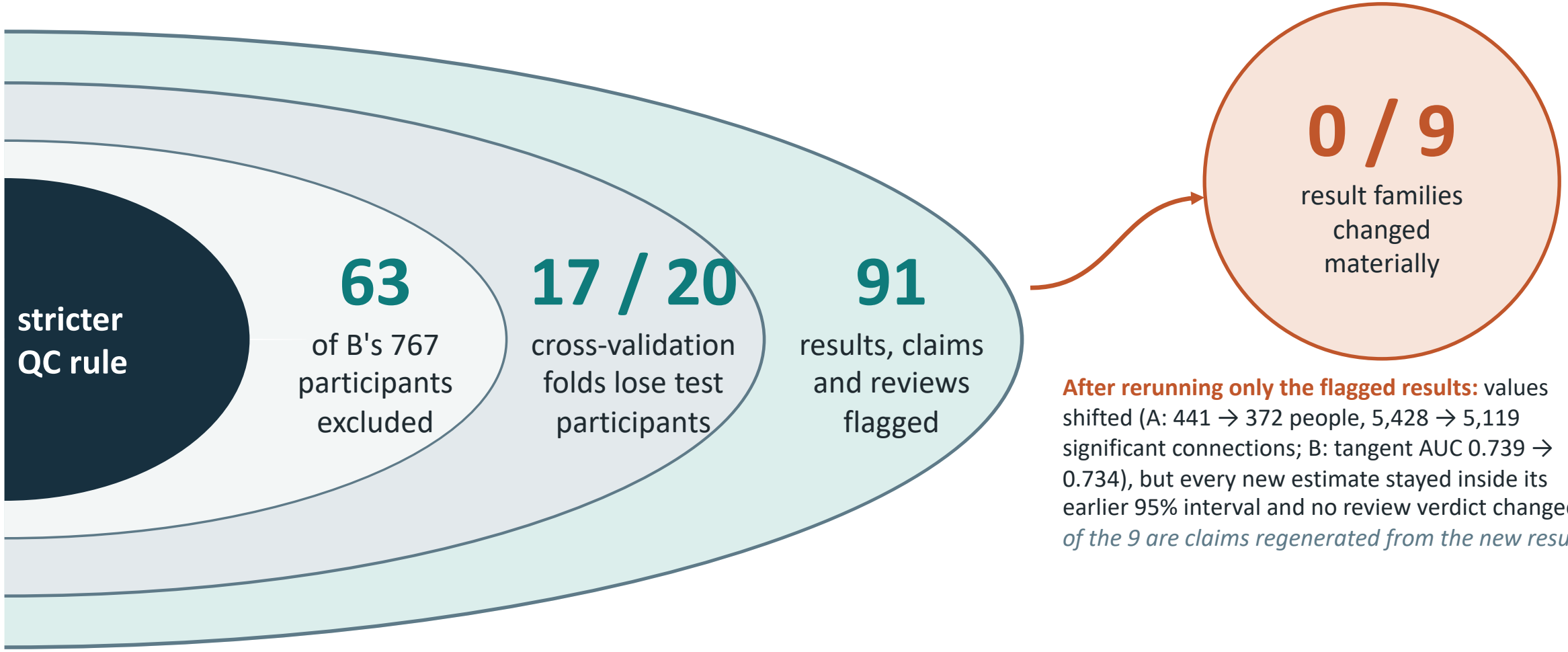
Before the change: B's cohort is 767 participants, tested in 20 leave-site-out folds; A's 441 participants all lie inside it.

1. How many of B's test participants are now excluded?
2. Which of B's results must be re-checked?
3. Which will actually change?

Write down your guesses.

Dependency tracing after a QC revision

In the graph: 4 SPARQL queries (queries/revision_*.rq) over graph/core_v1 before recomputation (code/revision.py); after rerun: results/revision_impact.json



After rerunning only the flagged results: values shifted (A: 441 → 372 people, 5,428 → 5,119 significant connections; B: tangent AUC 0.739 → 0.734), but every new estimate stayed inside its earlier 95% interval and no review verdict changed. *2 of the 9 are claims regenerated from the new results.*

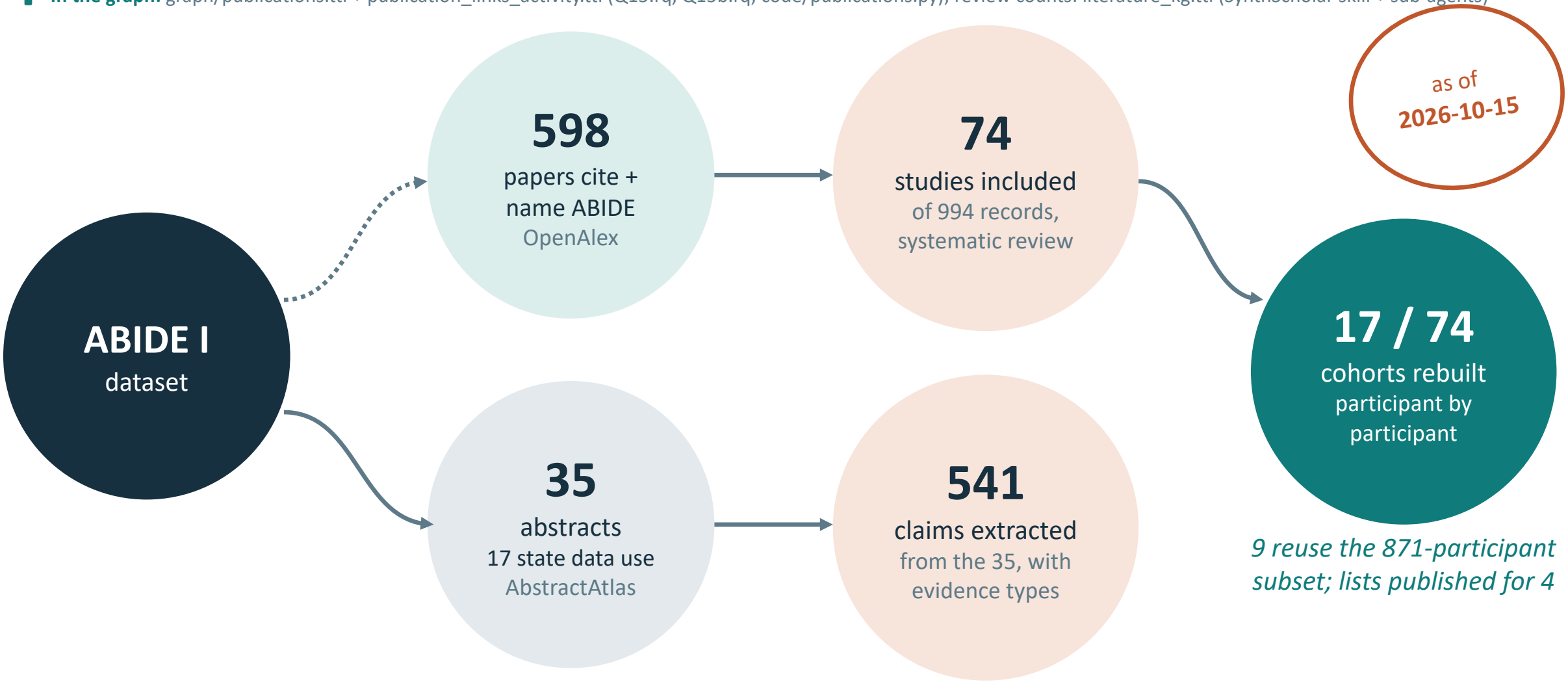
Flagged = depends on an excluded participant (traversal 0.05 s). **Changed materially** = new estimate outside the old 95% interval, or a review verdict changes. **Caveat:** the rule removes 23% of autistic but 14% of non-autistic participants, up to 88% at one site.

A *re-run · validated*

Data: ABIDE I (ABIDE Preprocessed, C-PAC)

Linking datasets, publications and abstracts

In the graph: graph/publications.ttl + publication_links_activity.ttl (Q15.rq, Q15b.rq; code/publications.py); review counts: literature_kg.ttl (SynthScholar skill + sub-agents)



A R S *re-run · validated*

Data: AbstractAtlas (OHBM 2026), OpenAlex; 74 ABIDE studies (systematic review, 2026-10-15)

Asking Claude about the literature graph

In the graph: literature_kg.ttl: ex:reportsPerformance, ex:evaluationScheme, ex:EvidenceQuote; query written and run by Claude Code, no skill

Participant asks

"Using only this graph: which studies tested their classifier by holding out whole sites, what accuracy did each report, and what verbatim quote supports it?"

88 s
Claude Code + pyoxigraph

Headless run in a fresh folder holding only literature_kg.ttl

Claude writes and runs SPARQL

```
# studies + accuracy (trimmed)
SELECT ?id ?acc ?basis ?snippet …
WHERE {
 { ?pub ex:reportsPerformance ?perf .
   ?perf ex:evaluationScheme
         "leave-site-out" }
 UNION { ?pub a ex:Publication ;
   ex:evaluationScheme
         "leave-site-out" }
 OPTIONAL {
   ?pub ex:reportsPerformance ?p2 .
   ?p2 ex:accuracyPct ?acc ;
       ex:basis ?basis }
}
# then the verbatim quotes
?q prov:wasQuotedFrom ?pub ;
   prov:value ?quote ;
   ex:sourceQuoteVerified ?verified .
```

Claude answers

Study	Leave-site-out accuracy
Abraham 2017	66.8%
S-030	56% balanced accuracy
S-062	71% ± 6%
S-071	73.4% (CC200 atlas)

Also in its table: S-072 ~62%; S-022 with no value

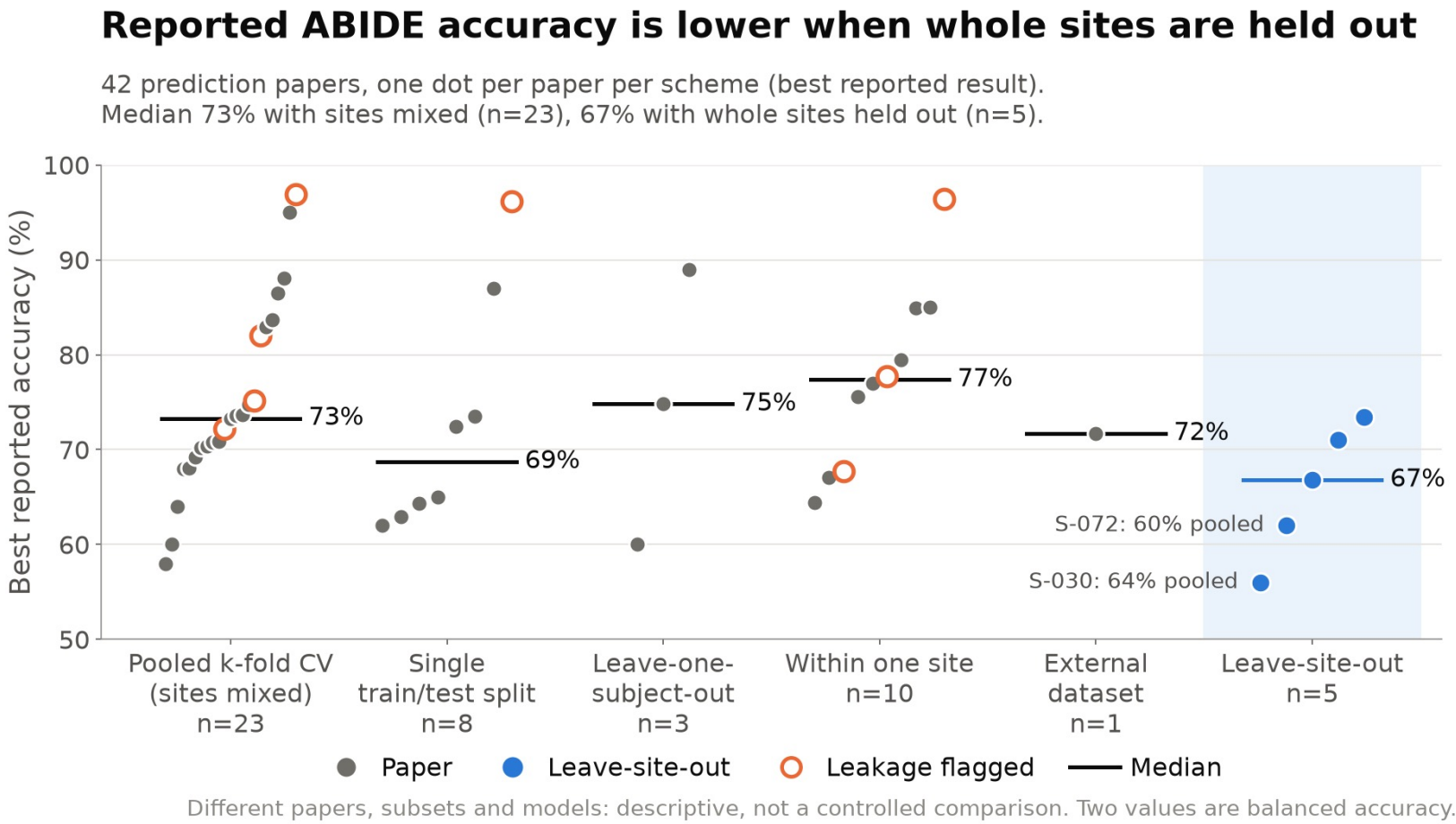
What the graph can't tell you

- S-022 ran the test, but the graph stores no accuracy for it
- Quotes are tied to papers, not to a specific accuracy value

A *validated* Data: 74 ABIDE studies (systematic review, 2026-10-15)

Evaluation schemes in 74 ABIDE studies

In the graph: literature_review/literature_kg.ttl (10,779 triples); phase3/s5_sparql_checks.py confirms each count → figure: phase3/s6_figures.py



Pooled k-fold

sites mixed across training and test folds

Leave-site-out

whole sites held out for testing

Descriptive, not controlled

different studies, models and subsets; best reported result per paper. Same model, both schemes: S-030 64% → 56% (balanced accuracy); S-072 ~60% → ~62%

How it was made

Skill: SynthScholar (protocol, retrieval, export); screening and charting by 8 sub-agents; each quoted value checked against full text, not by a person

Review: 994 records → 99 assessed → 74 included · of 53 with a predictive claim, 26 mix sites across folds and 6 hold whole sites out

S A R *validated* Data: 74 ABIDE studies (systematic review, 2026-10-15)

ReproPond and ReproLake as attributes on graph records

In the graph: graph/scope.ttl: knowledgeScope and accessLevel on 83 records; consentScope, egressPermitted on 3 profiles; 48 gate assessments (Q11.rq, Q12.rq)

Federated query (Neurobagel): the query goes to each ReproPond (a lab's local metadata store); only permitted results return.

Site ReproPond
scanner B quirk; motion profile

Lab ReproPond
FD rule, cohorts, reviews, skills

pond → lake, through gates

- generality across sites
- review
- consent covers broad sharing (DUO)
- licence
- persistent ID + version
- provenance

ReproLake
ReproNim's shared store of public dataset metadata

BIDS spec and conventions
published guidelines
dataset records, scoped claims
shared skills, variable definitions

Deprecation / revision

Stays in the pond

- consent scoped to study or named uses
- direct identifiers, full dates
- undefaced images
- quasi-identifiers, small cells
- restricted fields; linking keys

lake → pond: pin version, record local parameters, re-validate

Example: building a graph in Claude Code

In the graph: Skill: neuroimaging-kg (agent_skills PR #49), followed by Claude Code · example graph abide-caltech/graph/*.ttl, validated, loaded into Oxigraph

```
$ git clone https://github.com/sensein/agent_skills.git
$ cd agent_skills && git checkout neuroimaging-kg && cd ..
$ mkdir abide-kg-demo && cd abide-kg-demo
$ python ../agent_skills/scripts/install_skills.py --agent
claude --scope project --skills neuroimaging-kg
$ claude
```

Prompt 1

```
> Use the neuroimaging-kg skill to build a knowledge graph
for ABIDE I site Caltech — the phenotype table plus
ReproNim's NIDM file with brain volumes — validate it, and
load it into Oxigraph. Tell me in plain words what is in the
graph.
```

Prompt 2

```
> How many Caltech participants have both a full-scale IQ
(FIQ) score and a FreeSurfer hippocampal volume, by
diagnostic group? Show the SPARQL query you used and explain
it briefly.
```

Output of a test run, 2026-10-15

Prompt 1 · 34 s

"Done — the graph built, validated, and loaded cleanly, with all 5 gated questions passing."

"38 participants from the Caltech site of ABIDE I"

Prompt 2 · 14 s

"18 autism-group participants (dx_group=1) and 19 control-group participants (dx_group=2)"

Error in response 1: it said three tools produced the volumes; Caltech's file has two (FreeSurfer, FSL)

S L *validated* Data: ABIDE I, Caltech site (ABIDE Preprocessed; ReproNim simple2 NIDM, commit e0dff17)

Example: querying and using the graph

In the graph: Skill: neuroimaging-kg · SPARQL over the example store's named graphs (urn:g:phenotype, urn:g:repronim, urn:g:cohort), as in the tested transcript

Prompt 3

> Trace participant 51456's left-hippocampus volume from FreeSurfer back to the scan and the software it came from. Say which links are recorded in the graph and which are only inferred.

Prompt 4

> If we tightened the head-motion rule to mean framewise displacement below 0.15 mm, which participants would be excluded, and which results in our graph depend on them?

Output of a test run, 2026-10-15

Prompt 3 · 162 s

"fs:fs_003343 = 4513.9"

"T1w scan ↔ FreeSurfer activity … Not recorded."

Prompt 4 · 55 s

"51456 | autism | 0.1862 mm"

"51469 | autism | 0.1559 mm"

"17 autism / 19 control"

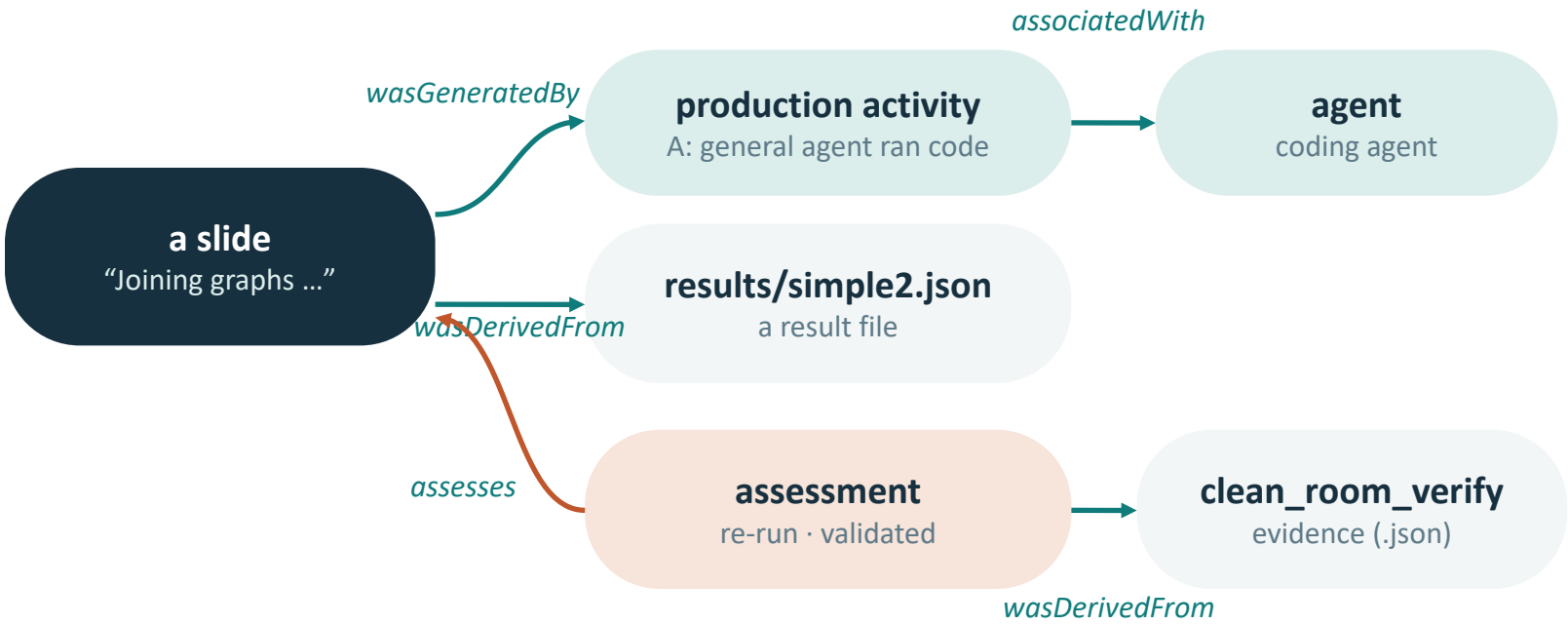
Gap in response 3: which scan FreeSurfer ran on is not recorded in the graph

S L *validated*

Data: ABIDE I, Caltech site (ABIDE Preprocessed; ReproNim simple2 NIDM, commit e0dff17)

The lecture's own provenance graph

In the graph: talk_provenance.ttl, written by slides/build_deck.js (pptx skill, deck-builder sub-agents); query run with pyoxigraph (prov_query.py)




```
SELECT ?status (COUNT(?slide) AS ?n)
WHERE { ?a talk:assesses ?slide ;
          talk:checkingLabel ?status
}
GROUP BY ?status
```

```
16  validated
16  re-run · validated
15  reviewed
6  unverified
3  re-run
2  re-run · reviewed
1  re-run (UMAP-derived values differ
slightly across machines)
1  re-run (UMAP differed ≤ 0.011)
```

Who produced it: speaker instructions → orchestrating Claude Code session → **general sub-agents (A)** built the experiments, deck and figures; **named skills (S)** only where stated: pptx (deck), SynthScholar (review retrieval), StructSense (claim pilot, Codex), neuroimaging-kg (examples, harmonisation); Brain Researcher via MCP (M) → checks → speaker review

Errors found by the speaker

- Consent wording: scoped consent, not “no consent”
- Kim 2023 question paraphrased wrongly; fixed from source
- AbstractAtlas links dropped, then restored

Errors found by automated checks

- Failed ORCD job: invalid p-values
- Clean rebuild caught a blocked download (HTTP 403)
- Two agents reconstructed the same cohorts differently

H L S A *reviewed*

Practices to adopt now

In the graph: Each practice adds a record: dataset version, cohort membership list, claim scope, derivation activity, shared identifiers

Objective 1: represent results as a graph

Objective 2: connect across graphs



Pin your data
commit or DataLad version, hashes

Save the manifest
included, excluded, and the rule

Scope each claim
population, measure, contrast, evaluation

Recompute one value
from source, before trusting a table

Use shared identifiers and terms
IDs for papers, people and tools; shared variable terms

Discussion

Which links between your lab's graph and others would you need before using work done with an AI assistant?

Packages with code, graphs and results:

CNP_Dataset_Publication_Knowledge_Graph.zip

ABIDE_Collaboration_Knowledge_Graph.zip

github.com/sensein/agent_skills · abstractatlas.brainkb.org

Poll: participant count in ds000030

259
A

94
C

255
E

290
B

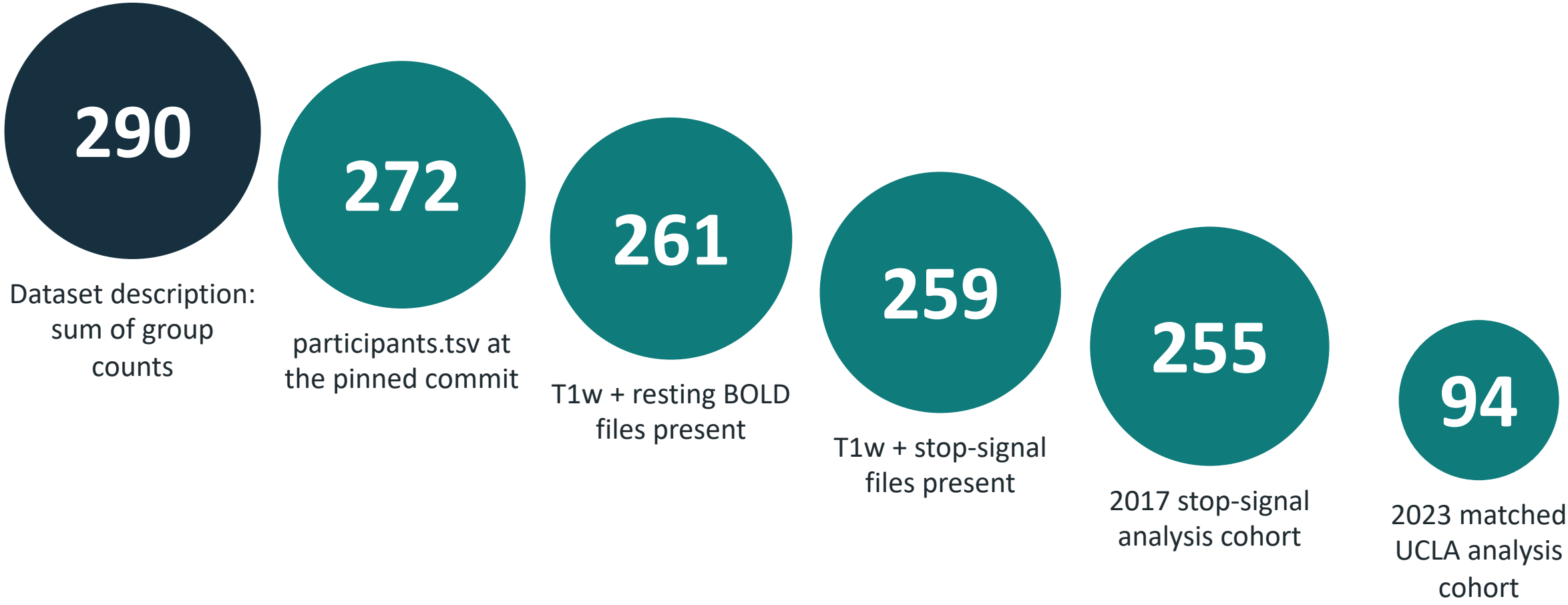
272
D

261
F

Hold up a letter, A–F

Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

Participant counts in ds000030 by scope



Data: ds000030 (UCLA CNP, commit 4070b6e); Gorgolewski 2017 and Kim 2023 cohorts

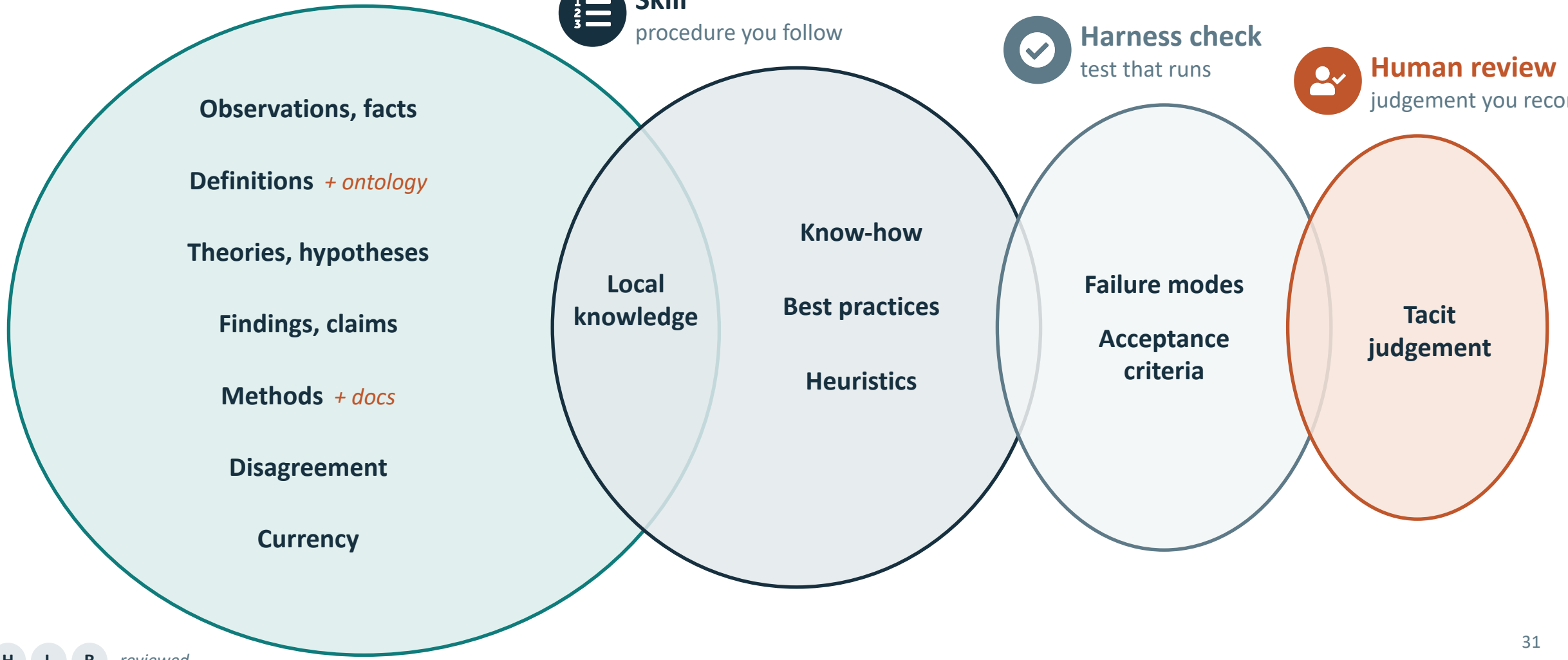
Types of knowledge and where they are recorded

Graph
records you query

Skill
procedure you follow

Harness check
test that runs

Human review
judgement you record



H L R *reviewed*

Recomputing a derived score from source items

sub-10159 · Barratt parcel 1



matches the reported value, 2

Dictionary formula: mean of items 19 and 17



reported scores reproduced from their item responses

P R *re-run* Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

Checking agent outputs against the graph

The skill, once per version

- held-out cases with human answers
- injected faults it must catch

Each output, every time

- check against the source
- keep cohort, direction, method

Faults we injected, and what caught them

Wrong offsets repaired and marked	**Invented quote** dropped	**Number as text** schema rejects	**Source edge removed** SHACL rejects

Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

Graph, search and Brain Researcher: a small comparison

Question	Our graph (gold)	Brain Researcher (MCP tools only)
Analyst B's QC cohort	767 (348 ASD, 419 TD)	Not answerable: one ABIDE node, 2,226 subjects
A–B overlap; excluded by stricter QC	441 (A within B); 63 excluded	Not answerable: neither paper in its graph
Trace one connectivity feature	Time series URL + hash, C-PAC settings, raw scan	Dataset level only: landing page, generic C-PAC node
Results to re-check after QC change	91 flagged by traversal; 0 of 9 changed	Not answerable
Does A's effect hold at held-out sites?	Not established (pooled d −0.08)	Not answerable: no matching nodes
Who reviewed what, on what evidence?	Six attributed assessments	Not answerable

Data: ABIDE I (ABIDE Preprocessed, C-PAC); Brain Researcher KG

Scope of the ds000030 extraction

Layer	Completed work
Phenotype tables	52 tables, 1,983 columns, 272 participants
Observed / missing cells	411,584 values / 127,792 missing cells
MRI inventory	2,531 file pointers, plus selected metadata
Papers	Three related papers, selected claims and methods
Graph size	834,327 NIDM triples before the claim bridge

Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

NIDM and PROV roles for each record

Record	NIDM / PROV representation
Participant and visit	prov:Person and nidm:Session
Questionnaire or scan	nidm:Acquisition activity
Variable definition and value	nidm:DataElement on an AcquisitionObject
Processing and its output	nidm:Derivative and DerivativeObject

Requirements for joining behaviour to image-derived measures

Link required	State in ds000030
Same participant and visit	Participant IDs available; time alignment unresolved
Imaging measure definition	Region, atlas, laterality, units and method required
Derived value to original image	Stats file, processing record and source image required
Example future join	SSRT with thickness or regional volume

Research questions and results of the three papers

Paper / result	Question, result and scope
Poldrack 2016	Data descriptor: enables dimensional study across diagnoses
Gorgolewski 2017	Reusable preprocessing and QC outputs
Kim 2023	Which direction of connectivity change is associated with schizophrenia; cross-dataset transfer
Kim 70.2%	UCLA, both directions, leave-one-out
Kim 61.5%	Train UCLA, test COBRE, decreased connectivity
Kim reverse transfer	No above-chance predictions reported

P R *validated* Data: ds000030 (UCLA CNP, commit 4070b6e); Poldrack 2016, Gorgolewski 2017, Kim 2023

StructSense extraction pilot

Stage	What actually ran
Contract	Frozen schema for claims, scope and evidence
Extraction	Host-model extraction of six bounded records
Alignment and review	Explicit vocabulary lookup and source review
Validation	Upstream span code, Ajv schema, RDF checks
Example record	Statement, scope, exact span + hash, open gap

Data: ds000030 (UCLA CNP, commit 4070b6e); Gorgolewski 2017, Kim 2023

Attribution across collaborators

Question the graph can answer	Required records and constraints
Who worked on this question?	Person → activity → question, claim and outputs
Which claims inform another question?	Reviewed relevance or support relation with scope
Who disagrees, and on what basis?	Separate assessments, evidence and reviewer roles
What changes when evidence is corrected?	Traverse affected claims, questions and work items
AbstractAtlas building blocks	Semantic search, extracted claims with ECO codes, author records, cached provenance

Design for a graph-versus-search comparison

Controlled factor	Proposed experiment
Same corpus, model and tool budget	Search only; skill only; graph only; both
Held-out questions	Single fact, cohort join, lineage and contradiction
Scoring	Correctness, evidence completeness, false claims
Longer agent sessions	Constraint violations and recovery

P *unverified*

Forms of drift and corresponding checks

Drift	Enforcement mechanism
Dataset or skill version changes	Pin and compare artifact hashes
A result loses its evaluation scope	Require claim-to-evaluation links
A derivative loses its source	Reject at the provenance gate
A report is treated as replication	Preserve evidence status and review changes
Still required	Image bytes, IDPs, paper manifests, held-out skill studies

Scan, events and phenotype details

Record (sub-10159)	Observed value
Resting BOLD TR / TE	2 s / 0.03 s, 3 T, scanner 35343
Stop-signal events	256 rows: 96 GO, 32 STOP, 128 NULL
STOP outcomes	17 successful, 15 unsuccessful
sst_ses_ssrt_quant / sst_ses_ssrt	200.125 / 216.625, unit not specified
T1w TR / TE: paper vs JSON	1.9 s / 2.26 ms vs 2.53 s / 3.31 ms

Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

Registration and cross-source joins

Item	What the evidence supports
Functional → T1	FreeSurfer BBR, 9 DOF (2017 paper)
T1 → MNI152NLin2009cAsym	ANTs SyN with affine initialization
Seven missing-T1 IDs	Same IDs in current metadata and 2017 Table 1
Kim 2023 and ds000030	Explicit accession and version link
Kim 2023 and 2017 derivatives	Reuse not established; different pipeline reported

Data: ds000030 (UCLA CNP, commit 4070b6e); Gorgolewski 2017

Skills and pull requests referenced

Skill / PR	Contribution
structsense	Entities and source-stated relations with evidence
synthscholar	Review protocols, corpus handling, retrieval provenance
labnb / duct / orcd-remote	Run records, resource logs, remote execution
sensein PRs #33, #34, #36	LinkML records, judge loop, MeSH search (open)
K-Dense datalad #227 / STAMPED #265	Retrieval provenance (merged) / evidence map (open)

Brain Researcher: reported benchmarks

Reported benchmark	Without BR	With BR
First-choice route/tool selection	23.3%	93.6%
Verified evidence grounding	4.6%	22.0%
Source	Chen et al., arXiv:2608.19902v1	preprint

Workshop connections

Earlier session	Reusable artifact this talk adds
Brent: LLM interaction	Task instructions with explicit inputs and outputs
JB: analytical flexibility	Named alternatives and comparable evaluation records
Michelle: automation	Validated inputs and inspectable tool execution
Yarik: STAMPED	Versioned components and evidence for replayability
Johanna: Project-Pilot	Reviewed changes to persistent skills and workflows

Sources

Source	Locator
ds000030 metadata, pinned	OpenNeuroDatasets/ds000030 @ 4070b6ee
Poldrack et al. 2016	doi 10.1038/sdata.2016.110
Gorgolewski et al. 2017 v2	doi 10.12688/f1000research.11964.2
Kim et al. 2023	doi 10.3389/fpsyt.2023.1232015
Skills and harness	sensein/agent_skills · K-Dense scientific-agent-skills · brain-researcher-public

Entities in a neuroimaging knowledge graph (1 of 2)

Group	Kind	Standards / examples
What was studied	Datasets and versions ●	BIDS, DataLad, OpenNeuro/DANDI/NDA accessions, DCAT
What was studied	Participants, phenotypes ●	NIDM-Experiment, ReproSchema, NIH CDEs, NDA data dictionary
What was studied	Acquisitions, protocols ●	BIDS sidecars, ReproIn, scanner/sequence parameters, ReproStim
What was studied	Tasks and constructs	Cognitive Atlas, CogPO, task events
How it was done	Methods and software ●	CodeMeta/schema.org, containers (ReproNim/containers, Neurodocker), Boutiques
How it was done	How to apply a method ●	Registration routes, compatible spaces, QC criteria, failure modes — skills with acceptance checks
How it was done	Templates and spaces	TemplateFlow (MNI152NLin2009cAsym, fsaverage), Julich/Allen, transform chains
How it was done	Run provenance ●	W3C PROV, NIDM, DataLad run records, ReproMan, BABS
How it was done	Compute environments	Container images, HPC profiles, seeds (reproseed)

H L *reviewed*

● *populated in the ds000030 or ABIDE packages*

Entities in a neuroimaging knowledge graph (2 of 2)

Group	Kind	Standards / examples
What was found	Derivatives and IDPs ●	BIDS-Derivatives, NIDM (FreeSurfer/ANTs/FSL → NIDM), MRIQC → NIDM
What was found	Statistical results, maps	NIDM-Results, NeuroVault, effect sizes with cohort
What was found	Questions, hypotheses ●	Preregistrations, research-question nodes
What was found	Claims, with scope ●	Population, measure, contrast, evaluation; ECO evidence codes; nanopublications
Who and in what context	Publications ●	DOI, PMID, sections and quotes as evidence locators
Who and in what context	People and roles ●	ORCID, ROR, CRediT, PROV qualified roles
Who and in what context	Projects and funding	NIH RePORTER grant IDs
Who and in what context	Reviews and disputes ●	Attributed assessments; who asserted what, on what evidence
Who and in what context	Resources and activity ●	Dated activity snapshots (commits, issues, citing papers) with age
Who and in what context	Ontologies ●	UBERON, InterLex/NeuroNames, NCIT, EDAM, ECO, SIO, nidm-terms
Who and in what context	Consent, use, licences	DUO, data use agreements, CC0/CC-BY

H L *reviewed*

● *populated in the ds000030 or ABIDE packages*

Types of knowledge and where they are recorded: full table

Kind	Neuroimaging example	Where it lives
Observations, facts	TR = 2 s for this scan; mean FD = 0.18 mm	Graph, with source
Definitions	SSRT computed by quantile method; RAS+; MNI152NLin2009cAsym	Graph + ontology terms
Theories, hypotheses	Dysconnectivity hypothesis of autism	Graph, as question/hypothesis with status
Findings, claims	Under-connectivity in ASD, in this cohort, this measure	Graph, scoped, with evidence
Methods	What tangent-space connectivity estimates	Graph + docs
Local knowledge	Scanner B's phase-encoding quirk; lab naming conventions	Graph + skill, scoped to site
Disagreement	Two reviewers disagree on QC	Graph, as separate attributed assessments
Currency	Tool last released 3 weeks ago	Graph, dated snapshot with age
Know-how	How to register BOLD → T1 → template and check it	Skill
Best practices	COBIDAS; preprocessing inside CV folds	Skill, citing the guideline
Heuristics	Exclude mean FD > 0.2 mm	Skill parameter, with its derivation and limits
Failure modes	Left–right flips; leakage across folds; site confounds; double dipping	Harness check / test
Acceptance criteria	Recomputed score matches; derivative has a source edge	Harness check (SHACL, tests)
Tacit judgement	Visual QC of a registration	Human review, recorded as an attributed assessment

ReproPond and ReproLake: full table and transitions

ReproPond (lab / site)	ReproLake (shared)
Scanner B phase-encoding quirk; site motion profile	BIDS spec, scanner-independent conventions
Lab QC threshold and its derivation (FD < 0.2 mm on our data)	Published guideline / consensus (COBIDAS, OHBM best practices)
Exact cohort manifests, exclusions, per-participant derivatives	Dataset record, version, licence (OpenNeuro, DANDI, ABIDE)
Lab skills and runbooks (how we register, how we QC)	Shared skill repos (ReproNim/skills, sensein/agent_skills)
Local variable names, site-specific instruments	nidm-terms / InterLex concepts, ReproSchema library
Unpublished findings, reviews, disagreements	Publications, scoped claims in literature KGs
Data with consent scoped to the study or named uses; DUA-limited data	DUO codes; only de-identified, permitted summaries

pond → lake: generality beyond one site or sample, review, de-identification, consent that covers broad sharing (DUO), licence, persistent identifier and version, provenance retained.

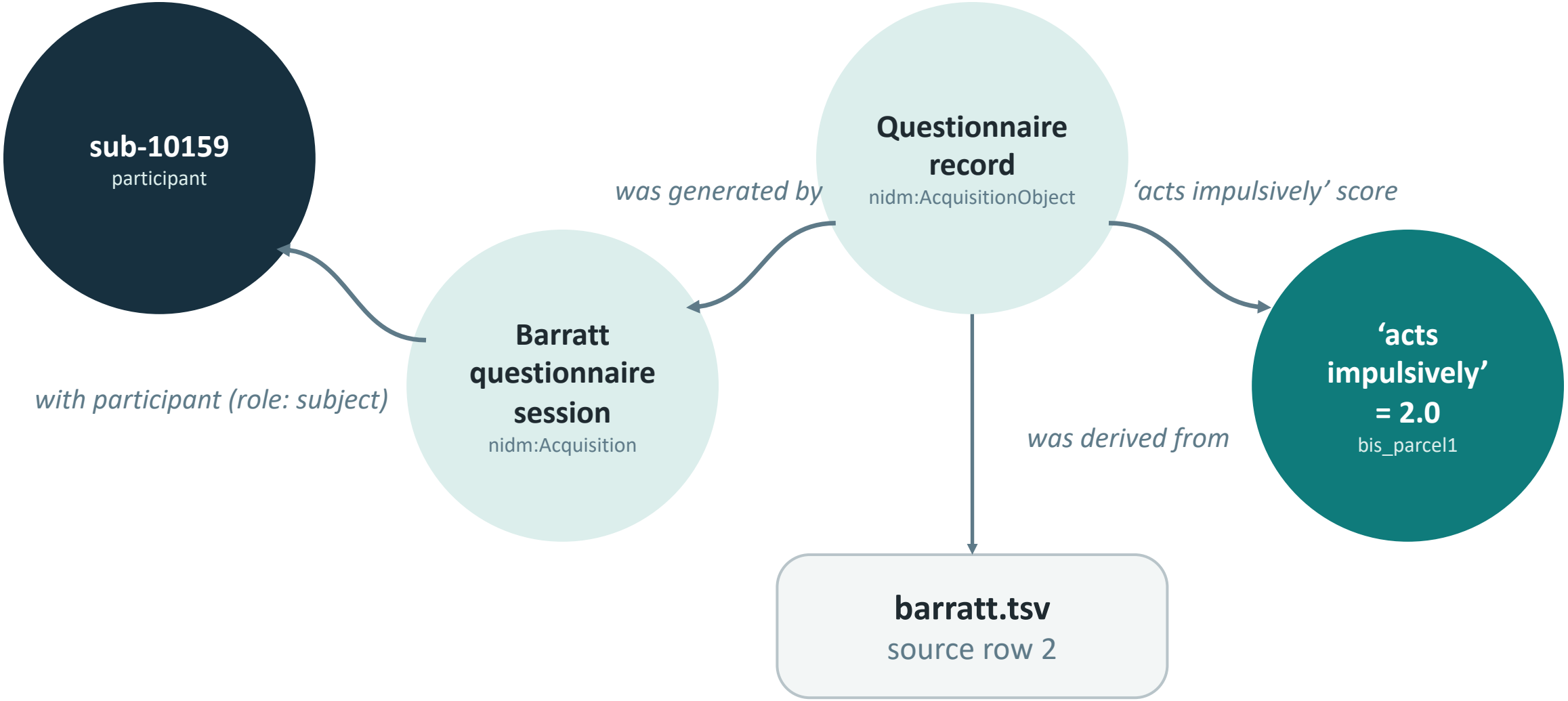
lake → pond: pin the version, record local parameters and deviations, re-validate on local data.

Deprecation / revision: shared knowledge ages or is contradicted; flag local copies when the source changes.

Stays in the pond: data whose consent does not cover broad sharing (consent scoped to the study or to named uses; DUO-restricted; DUA-limited); direct identifiers and full dates; undefaced structural images; quasi-identifier combinations and small cells; restricted fields; linking keys between pseudonymous IDs and people.

Can flow after the gates: dataset-level metadata, dictionaries and term mappings, aggregate counts above a minimum cell size, de-identified consented records, derived measures where consent allows, skills, provenance with pseudonymous IDs, scoped claims.

Provenance of a single phenotype measurement



Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

AbstractAtlas: building blocks

Available building block	Pointer for this workflow
Explore abstracts and related literature	Semantic search, facets and NeuroScape map
Inspect extracted claims	Source text, evidence category and ECO annotations
Connect authors to abstracts	Author records and accepted poster identifiers
Preserve computation history	Source snapshots, model IDs and component caches

abstractatlas.brainkb.org · github.com/sensein/abstractatlas

Resource discovery with dated activity records

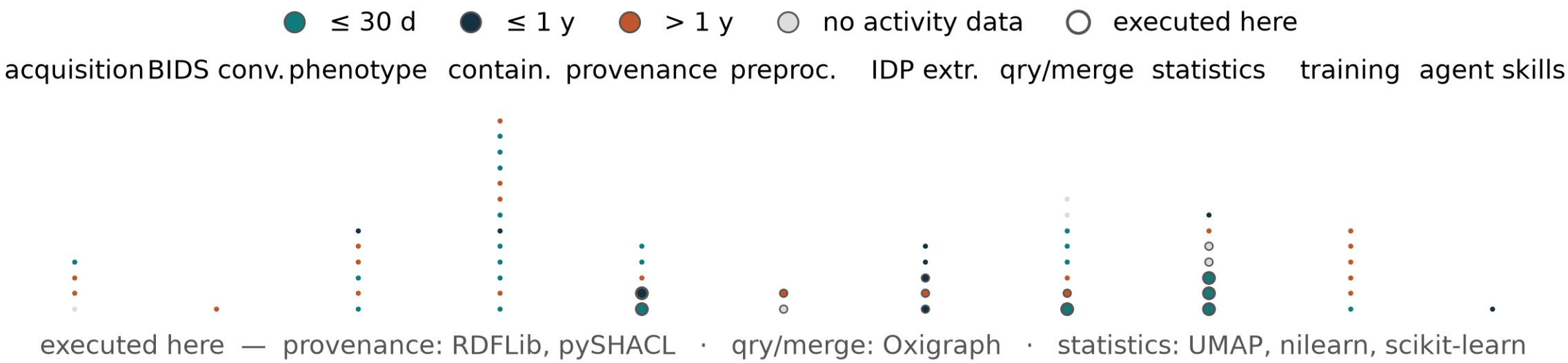
Query “convert FreeSurfer stats to NIDM”

snapshot 2026-10-15 14:57 UTC, 0.46 h old at query time

1 **FreeSurfer NIDM BIDS App**
IDP extraction to NIDM
13 commits in 90 days, release v0.1.1
last commit 2 days before snapshot

2 **segstats_jsonld**
IDP extraction to NIDM
0 commits in 90 days, used here as data source
last commit 318 days before snapshot

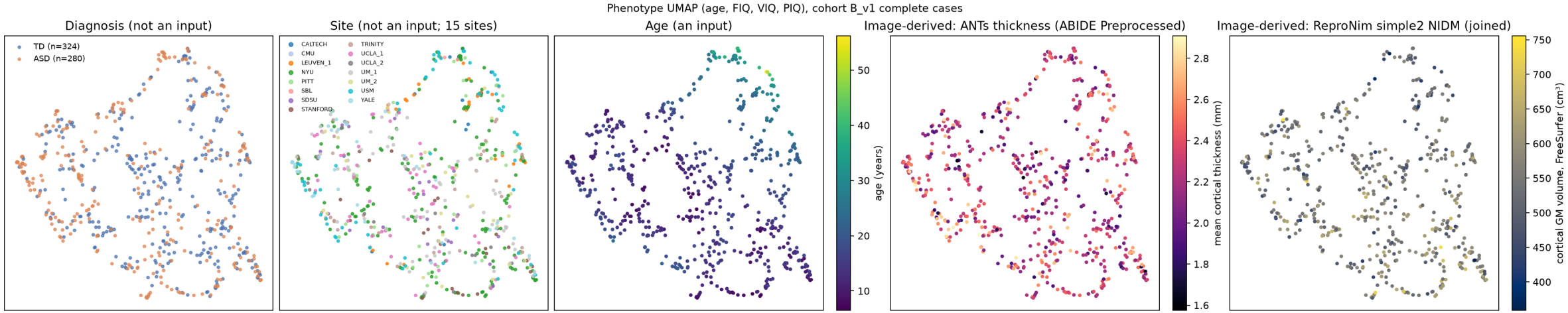
3 **MRIQC NIDM BIDS App**
QC to NIDM
10 commits in 90 days, release v0.1.1
last commit 2 days before snapshot



A R *re-run · validated*

Phenotype embedding linked to image-derived phenotypes

Backup



- 604 of B's 767 with age, FIQ, VIQ and PIQ; 15 sites
- trustworthiness 0.977 (PCA 0.920); other seeds 0.973, 0.974
- silhouette by diagnosis 0.03, by site −0.34

A *re-run (UMAP-derived values differ slightly across machines)*

Data: ABIDE I (ABIDE Preprocessed, C-PAC)

Glossary

ds000030: UCLA Consortium for Neuropsychiatric Phenomics dataset (OpenNeuro): schizophrenia, bipolar disorder, ADHD and controls

ABIDE I: Autism Brain Imaging Data Exchange: resting-state fMRI and phenotypes, 17 sites, 1,112 participants

ASD / TD: autistic / typically developing comparison participants

Barratt scale (BIS): 30-item impulsivity questionnaire; items scored 1 rarely to 4 almost always

bis_parcel1: 'acts impulsively': mean of items 17 and 19

SSRT: stop-signal reaction time: how long it takes to cancel a started response

FIQ · SRS · ADOS: full-scale IQ · Social Responsiveness Scale · Autism Diagnostic Observation Schedule

Mean FD: average head movement between fMRI volumes, in mm

C-PAC · fMRIPrep: preprocessing pipelines (ABIDE Preprocessed used C-PAC)

CC200: 200-region functional brain parcellation; connectivity = correlation between regions

BBR · ANTs SyN: fMRI-to-anatomy registration · non-linear registration to a template

CPM · linear SVM: connectome-based predictive modelling · a standard linear classifier

LOOCV · leave-site-out: leave one participant out · train on all sites but one, test on the held-out site

AUC · FDR · d: 0.5 = chance, 1 = perfect · false discovery rate correction · standardised effect size

UMAP · trustworthiness: 2-D map placing similar participants close · how well neighbours are preserved

Triple · query: one statement: subject, relation, object · a statement pattern with blanks

RDF · SPARQL · Oxigraph: file format for statements · query language · a graph database

NIDM · PROV: neuroimaging data vocabulary · provenance standard (what produced what, from what)

SHACL: machine-checkable rules a graph must satisfy

UBERON · InterLex · ECO: anatomy vocabulary · shared variable definitions · evidence types for claims

DUO · DOI · ORCID · RRID: data-use conditions · persistent IDs for papers, people, tools

PROBAST · ComBat: risk-of-bias tool for prediction studies · site-effect harmonisation

ReproPond · ReproLake: a lab's local metadata store · shared store of public datasets

Neurobagel · MCP: sends a query to each pond, returns only permitted results · plug-in interface for AI tools

Key to the AI-involvement badges

How it was produced

- **H** — **Speaker:** written or decided by the speaker
- **L** — **LLM draft:** drafted by an LLM from the speaker's instructions, then edited by the speaker
- **S** — **Named skill:** an agent followed a named skill (named on the slide: SynthScholar, StructSense, neuroimaging-kg, pptx)
- **A** — **General agent work:** Claude Code or its sub-agents writing and running their own code (Python, SPARQL, Slurm on ORCD), no skill
- **R** — **Agent retrieved:** retrieved by an agent from cited sources; quotes checked against the source
- **M** — **External agent (MCP):** answered by an external agent system through MCP (Brain Researcher)
- **P** — **Earlier agentic session:** produced in an earlier session by a different agentic system, reused

Per-slide records are also exported as W3C PROV in talk_provenance.ttl.

How it was checked

re-run
recomputed independently (for example a clean rebuild on ORCD matched)

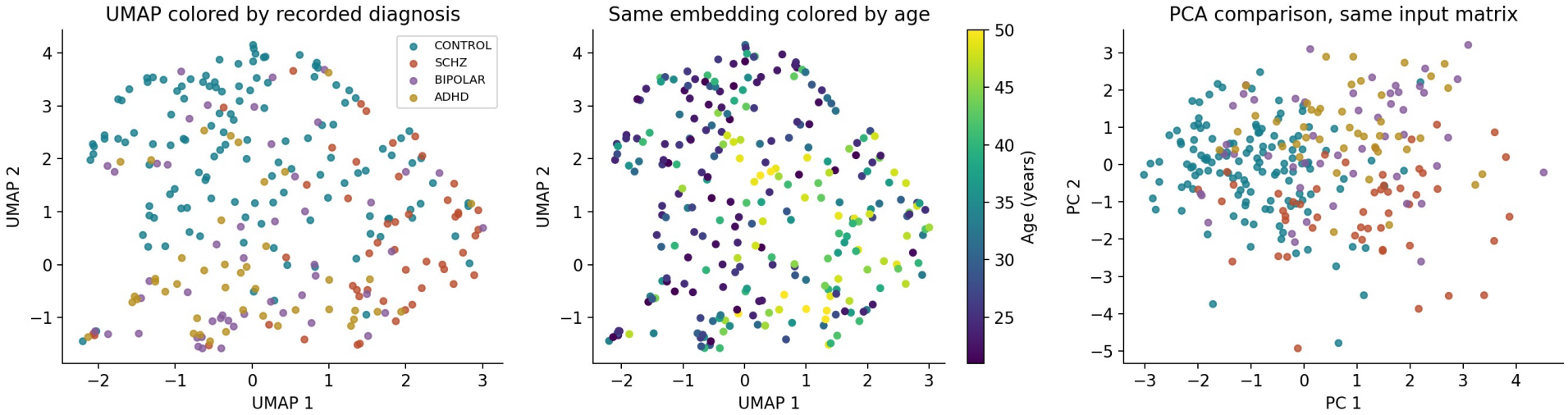
validated
passed automated checks (SHACL, schema, tests, render QA)

reviewed
read and corrected by the speaker

unverified
not independently checked; stated as such

ds000030 phenotype embedding from a graph query

272 participants, 10 behavioral/cognitive scores; diagnosis excluded from inputs



Check	Original	Re-run (ORCD)
UMAP trustworthiness, k = 15, seed 42	0.874	0.879
PCA trustworthiness, same matrix	0.805	0.805
UMAP seeds 43 / 44	0.868 / 0.879	0.878 / 0.870
UMAP with 30 neighbours	0.867	0.860
15-NN overlap with seed 42 (seeds 43 / 44)	0.74 / 0.74	0.75 / 0.73

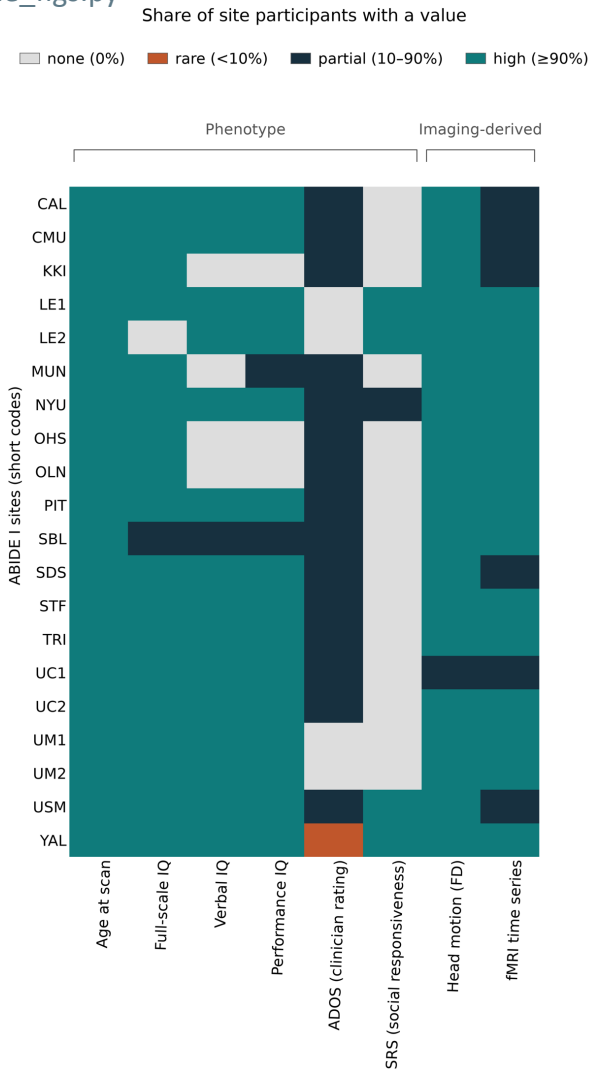
P *re-run (UMAP differed ≤ 0.011)*

Data: ds000030 (UCLA CNP), OpenNeuro, commit 4070b6e

Site coverage of each data element (ABIDE I)

Backup

In the graph: SPARQL (queries/Q14.rq) over graph/core + view/volumes + ReproNim simple2 graphs → counts by site and diagnosis (code/element_coverage.py → results/element_coverage.json) → figure: figs_slide/make_slide_figs.py



A *validated*

Data: ABIDE I